

technology empowers the consumers through two-way communication with the utility companies. Customer experience is a top business priority for any business. Repetitive type of customer-side tasks like payments, basic inquiries related to billing, complaints, changes in customer details, and the like can be simply solved through virtual agents and chatbots, saving time and money.

In rural areas especially, invoices and other records are still in handwritten form. With capabilities like natural language processing, AI can help in digitising these records. We can even go as far as instructing users on how to install or activate products related to energy management via AI

and computer vision if they want to set them up in their homes.

The result is better integration of large-scale renewable energy systems, thereby lowering peak demand, efficient power transmission, reduced operations, and management costs, which are essential to meet economic and environmental needs.

In 2018, SLAC National Accelerator Laboratory at Stanford University came up with a software platform called VADER, which stands for Visualization and Analytics of Distribution Systems with Deep Penetration of Distributed Energy Resources. It can model potential changes in connectivity and behaviour of various resources on the grid using AI for resource opti-

misation. In India, numerous startups like Avrioenergy, The Solar Labs, and many more are working in this sector.

While the training speed of algorithms has improved due to better computational power, AI presents threats like cyberattacks on an automated power grid. Applying AI in the right way is crucial to successfully deploy it on a large scale. Inadequate quality of training data can lead to false predictions. This causes a lot of time and effort to be spent on data categorisation and integration rather than the task at hand.

But given the potential benefits, as technologies become more advanced, there will be enormous adoption of AI worldwide in the energy sector.

—Ayushee Sharma

2 SMART EDUCATION TO IMPROVE SKILLSETS OF STUDENTS

Smart education is playing an important role in the education sector. It has helped in improving the skillsets of students and made learning more interactive and engaging

A Greek philosopher once said, “The only constant in life is change.” The same goes for technology, which has become extremely advanced in the past few years and brought many changes in our daily lives. It has also modernised various sectors, including education and learning. Information and Communication Technology (ICT) has played a key role in changing the dynamics of modern-day education, teaching, and learning.

ICT is deeply integrated into classrooms and coaching centres and has completely changed the traditional methods of teaching. It has also improved the skillsets of students such as global awareness, communication and collaboration, critical thinking and problem solving, social and cross-cultural skills, and self-direction.

These days learning is not only about gaining knowledge but also developing intrapersonal and cognitive skills. As rightly said by experts, the efficacy and success of learning depend not only on the outside environment but also on the suitability of the learning process.

Smart learning is nothing but an amalgamation of technologies.

It includes three imperative parameters—learning outcomes, learning behaviour, and mental systems.

The paradigm shift in education.

Smart education offers a huge paradigm shift in the way students access education. The rapid changes in technology make it difficult for teachers to understand what the future will be like in the coming years. Smart education solves this quandary with the use of cutting-edge technologies that help both teachers and students to prepare themselves for the road ahead. The unique aspect is that it can be done in both physical and virtual environments.

While speaking about smart education and its benefits in the coming years, we need to understand the importance of technology in education and how it is going to shape the future. The recent market research report from Technavio predicts that the smart classroom market in India will grow at a CAGR of more than twenty per cent in 2020. College, coaching, and school classrooms are now furnished with state-of-the-art tech-based education tools that completely wipe out the traditional methods of education. Electronic items such as projectors and document cameras have made colleges

and schools smarter and eco-friendly.

Benefits of smart education. While speaking about various aspects of smart education, we need to look at the various benefits it offers to the education sector. This sector has already adopted tech-based teaching methods with the use of interactive intelligent boards, projectors, smart notebooks, etc. No doubt, these gadgets make teaching sessions much more interactive and engaging. Let’s find out some of the top benefits of smart education in the education sector.

Interactive sessions. Students’ engagement rate has escalated by 55 per cent globally after the introduction of technology in schools and colleges, as per a report by Gallup. Instead of traditional blackboards, smartboards are now being used in schools to help teachers and college professors give lectures more effectively by utilising various projections. In addition, teachers can explain their chapters more easily with the help of special effects and graphics.

Role of online resources. A smart classroom is furnished with projectors and digital display boards that help teachers to provide practical solutions from the Internet in a trouble-free manner. Students can also look for